

CDSFL

Mathematical review suite

Reading copy | Proposal 1.1 | 10 September 2026

Prepared for George Jackson and a possible external review panel.

Status: proposed revision. Adoption and empirical validation remain open.

This reading copy contains the complete current proposal, lineage note, validation plan, preceding critique, and panel review materials.

The ZIP also preserves editable Markdown, exact executable checks, recorded results, source snapshots, review notes, and revision history.

For independent review, begin with `START_HERE.md` and the panel brief. Read the frozen original model before the preceding critique.

Equations and explanatory content are retained in the reading copy. The archived Markdown files remain the authoritative editable versions.

A correct conditional derivation does not establish calibrated inputs, practical superiority, universal applicability, or historical novelty.

Contents follow in this order:

1. Revised mathematical specification
2. Lineage and decay clarification
3. Experimental validation and falsification plan
4. Preceding mathematical review
5. Panel review brief
6. Panel response template

CDSFL: a revised mathematical core for evidence, revision, and scientific enquiry

Version 1.1 proposal - 10 September 2026 (London). Prepared for George Jackson. This is a completed mathematical revision proposal, with exact checks and a prospective validation plan. It is not a deployment, an empirical certification, or a claim of a new probability theorem.

Assessment: CDSFL can retain a compact, human-comprehensible core while correcting the observation/repair confusion and the coupling of publication novelty to evidential strength. The resulting core is mathematically sound under explicit definitions. Its accuracy as an operational scientific instrument depends on separately testable observation models, action models, and parameter estimates.

The source examined is Constraint Engineering at commit 9e3dfe9d68b428acb98e65c0ad3800ec571a4da4. The [preceding review](#) records the original objections. This proposal addresses the central coverage, residual-risk, repair, and novelty interpretation; it does not certify every cognitive or emergence metric elsewhere in the project. The work record (archive: outputs/CDSFL_revision_plan_and_log.md) preserves decisions and corrections.

1. The human-comprehensible collapsed form

Human-comprehensible account of the whole framework:

Capable repeated examinations progressively reduce the opportunity for an existing error to remain undetected. Their returns can diminish. Repairs can remove errors and introduce others. Discovery curves help us understand whether further investigation is productive.

The first sentence concerns an unchanged target and a consistently defined error. Each further examination must retain some ability to detect an error that earlier examinations could have missed; duplication alone need not add coverage. If the target or criterion changes, the revised state must be assessed explicitly.

The mathematical account can be stated and assessed independently of the CDSFL harness. Its definitions, conditional derivations, testable observation models, and empirical predictions apply to appropriately specified human, instrument, or computational investigations. CDSFL is one implementation and possible experimental vehicle for that account.

The standalone contribution proposed here is a framework relating repeated attempted refutation, corroboration, intervention, and the productivity of enquiry. Its mathematical ingredients include established probability laws. Whether their organization and operationalization make a distinct and useful scholarly contribution requires comparison with prior work and tests against external outcomes; the existence of software neither establishes nor is required for that assessment.

For a stated failure criterion on an identified target:

$$R_{\text{next}} = \frac{(1-s)LR + b(1-R)}{LR + (1-R)}$$

Symbol	Meaning
R	Probability that the current target fails the stated criterion, before the new observation.
L	How much more likely the actual observation would be if that failure existed than if it did not.
s	Probability that the chosen action leaves an initially failing target free of the specified failure, accounting for all effects of that action.
b	Probability that the action leaves an initially non-failing target with the specified failure.

Plain reading: use the evidence to reassess the risk, then account for failures the action removes and failures it introduces.

The same equation can be taught in two steps:

$$B = \frac{LR}{LR + 1 - R}, \quad R_{\text{next}} = (1 - s)B + b(1 - B).$$

B is the assessed risk after observing the evidence and before changing the target. Evidence with $L < 1$ weighs against the failure; $L > 1$ weighs for it; $L = 1$ leaves the odds unchanged. An observation alone uses $s = b = 0$.

The word “failure” is deliberately scoped. It can mean an invalid step in a derivation, a numerical routine violating a defined property, or a design failing a stated performance requirement. “Free of the specified failure” does not mean flawless in every conceivable respect.

This gives the wider project a defensible organizing principle: **enquiry should change our beliefs according to discriminating evidence; revisions should be assessed according to their effects; further work should be chosen according to its expected scientific and practical value.** These are connected operations with different mathematical meanings.

2. Definitions that must survive the collapse

Let H denote the specified pre-action failure event. Let $\mathcal{J}$ contain the target version, criterion, relevant history, test-selection procedure, evidence provenance, and assumptions. Then $R = P(H | \mathcal{J})$.

For the actual observation e , define likelihoods

$$\ell_1 = P(e | H, \mathcal{J}), \quad \ell_0 = P(e | \neg H, \mathcal{J}).$$

For continuous observations these may be densities against the same reference measure. The canonical formula is

$$R_{\text{next}} = \frac{(1 - s)\ell_1 R + b\ell_0(1 - R)}{\ell_1 R + \ell_0(1 - R)}.$$

The ratio form uses $L = \ell_1 / \ell_0$ when $\ell_0 > 0$. The likelihood-pair form also handles decisive evidence without representing an infinite ratio. Require a positive denominator. A zero denominator makes this conditional update undefined. For discrete outcomes, the supplied model assigns the observation zero probability; with densities, the representation and null-set conventions also matter. Record the issue and examine the model/data rather than substituting a clipped posterior.

The selected action a produces a new target version, whose failure event is H' :

$$s = P(\neg H' | H, e, \mathcal{J}, a), \quad b = P(H' | \neg H, e, \mathcal{J}, a).$$

Three conditions are essential:

1. **The event remains clear.** Repairing a proof produces a new proof. It does not make the old invalid inference valid. Revising a scientific hypothesis creates a different hypothesis; it does not change the truth of the original claim. Changed criteria require an explicit mapping or a new assessment.
2. **Action selection is accounted for.** The policy must choose a from recorded information, allowing randomization independent of the hidden state conditional on that information. If a person or model chooses using additional private evidence, incorporate that evidence or its selection mechanism into B . Observational repair rates do not automatically identify the effects of a different intervention policy.
3. **The probabilities share their conditioning.** s and b apply to this evidence, context, and action. An unconditional success rate from unrelated tasks is not automatically appropriate.

For an “any flaw in class K ” event, s means removal of **all** failures needed to make that event false. Fixing one reported issue does not establish this probability. Binary risk also omits flaw count, severity, and unlisted failure classes.

Conditional transition entries on zero-weight branches need not be estimated: any valid kernel value there gives the same result. A zero total observation denominator still makes the update undefined.

The core is a compact update interface to a potentially richer model. A single R does not contain the history needed to supply its next likelihoods. The claimed simplicity is in the arithmetic and explanation, not in pretending all relevant evidence fits into one number.

3. Derivation and boundaries

Bayes' theorem gives $B = P(H|e, \mathcal{J})$. Under the action-selection condition, choosing a adds no omitted information about H . Partitioning the post-action event over the two pre-action states gives

$$\begin{aligned} P(H'|e, \mathcal{J}, a) &= P(H'|H, e, \mathcal{J}, a)B \\ &\quad + P(H'|\neg H, e, \mathcal{J}, a)(1 - B) \\ &= (1 - s)B + b(1 - B). \end{aligned}$$

Substituting Bayes' expression yields the collapsed equation. This is an application of established probability laws. NIST describes the underlying prior-likelihood-posterior calculation in its [Bayesian reliability handbook](#).

Consequences follow directly:

- For $R, s, b \in [0, 1]$, nonnegative likelihoods, and positive denominator, the answer lies in $[0, 1]$: it is a weighted average of $1 - s$ and b .
- No action recovers Bayes exactly. No action and uninformative evidence leave R unchanged.
- A confirmed failure gives $B = 1$, hence $R_{\text{next}} = 1 - s$. Confirmed absence gives $B = 0$, hence $R_{\text{next}} = b$.
- A net perfect action $s = 1, b = 0$ leaves zero risk for the specified event.
- There is no requirement that $s + b \leq 1$. An action can cure flawed targets and damage clean ones. With $s = b = 1$, the state flips; the equation remains valid.
- Certain priors cannot be overturned by ordinary finite likelihood ratios. Reserve exact zero and one for declared logical or modelling commitments, not small samples or high confidence.
- Neither risk nor the marginal improvement must decline on every pass. Evidence can correctly reveal greater risk.

The no-action reductions assume the target remains unchanged during the interval. Environmental change requires its own transition even when no repair is attempted.

Successive observations on an unchanged target use **history-conditional** likelihoods. Their ratios multiply by the chain rule. Independence is a sufficient simplification; it is not a consequence of different model names, architectures, or prompts.

4. Observation and repair: worked cases

Let a binary review have sensitivity $q = P(D|H, \mathcal{F})$ and false-positive rate $f = P(D|\neg H, \mathcal{F})$. The actual observed branches give

$$B_+ = \frac{Rq}{Rq + (1-R)f}, \quad B_- = \frac{R(1-q)}{R(1-q) + (1-R)(1-f)}.$$

Timeout, abstention, and unresolved output require separate observation categories when they occur; they are not automatically negative findings.

With $f = 0$, the negative branch becomes

$$B_- = \frac{R(1-q)}{1-qR},$$

preserving the valid original CDSFL recurrence. It describes a clean review of the old target.

The repair counterexample, resolved

Assume one scoped flaw, $R = 1/2$, $q = 4/5$, no false positives, and perfect repair only after a positive result.

Outcome	Branch probability	Risk after observation	Risk after the prescribed action
Positive	2 / 5	1	0
Negative	3 / 5	1 / 6	1 / 6

The expected post-cycle risk is 1 / 10. The negative-branch risk is 1 / 6. The positive branch after its perfect repair has risk zero. The revised model preserves all three quantities without conflating them.

A clarification to the preceding review: zero remaining risk for a confirmed flaw assumes the perfect repair actually occurs. If an additional detector still gates repair with sensitivity q , the ex-ante risk is instead $1 - q$. The earlier recurrence returning one is incorrect under either of those repair policies when $q > 0$.

For general result-dependent actions with parameters (s_+, b_+) and (s_-, b_-) , enumeration gives

$$\mathbb{E}[R_{\text{next}}] = R\{q(1 - s_+) + (1 - q)(1 - s_-)\} + (1 - R)\{fb_+ + (1 - f)b_-\}.$$

With observation alone, $\mathbb{E}[B] = R$. Learning does not manufacture an expected reduction in the probability of a fixed proposition.

Repair followed by new errors

To retain the project's repair/reinjection vocabulary, suppose raw repair removes the specified failure with probability σ , followed by introduction probability ν on each intermediate clean path. Assume that conditional introduction rate is the same for initially clean and successfully repaired paths. Then

$$s = \sigma(1 - \nu), \quad b = \nu, \quad R_{\text{next}} = \nu + (1 - \nu)(1 - \sigma)B.$$

Raw efficacy σ and net removal s must not be interchanged. Different introduction rates on those paths can instead use $s = \sigma(1 - \nu_1)$, $b = \nu_0$.

The action schedule matters. In the numerical example above, let $\nu = 0.1$. New errors possible only on the positive-result repair branches give expected risk 0.14. An additional process that can introduce errors after every cycle gives 0.19. Neither schedule should silently stand in for the other.

After an action, verification supplies evidence about the **new** target. Passing the tests used to optimize the repair is not automatically independent confirmation.

Dependence, false alarms, and replication

The executable checks also establish these constructed examples:

Case	Correct result
Prior 0.1, sensitivity 0.8, false-positive rate 0.2, positive result	Posterior 4 / 13, not certainty.
Prior 1 / 2, a negative $q = 0.8, f = 0$ review	Posterior 1 / 6.
Copy exactly that already recorded observation on the unchanged target	Still 1 / 6: conditional likelihood ratio is one.
Collect a genuinely conditionally independent replication with the same operating characteristics, also negative	Posterior 1 / 26. Publication originality is irrelevant to this calculation.

A tool rerun on a changed target may supply new evidence; the duplicate rule concerns the same already-conditioned information. Literature provenance and publication-selection effects can legitimately affect an observation model. A generic novelty score is not a substitute for that model.

5. Preserve the valid lineage, correct the interpretations

Existing component	Proposed treatment
Coverage $C = 1 - \prod_i (1 - q_i)$	Retain with $q_i = P(D_i H, \text{all earlier misses, relevant context})$ on supported histories. Replacing conditional sensitivities by marginal ones requires a justified factorization; conditional independence given the flaw is sufficient.
Weighted class coverage F	Retain as a defined process measure. It is not automatically a fraction of all possible scientific failure modes. Estimated coverage needs calibration too.
Bayesian residual risk R	Retain, add false-positive and other observation channels, and state the event and conditioning explicitly.
Three-phase repair extension	Replace its repair interpretation with the observation-plus-transition derivation above. Do not preserve an incorrect reduction merely for compatibility.
Literature novelty and search coverage	Report separately, with scoped searches and provenance. They may influence discovery priorities; they do not automatically suppress corroborating evidence.
Diversity discounts and Ising dependence	Keep as candidate fitted approximations where useful. Pairwise structure is a restricted family, not a representation of all dependence.
Human/machine combination	Use conditional incremental detection. For detections M, J , $P(M \cup J H) = P(M H) + P(\neg M H)P(J \neg M, H)$. Retain other context throughout.
Tool gates and solution scores	Retain auditable admissibility checks. A bounded score or fraction of chosen tests passed is not automatically a calibrated sensitivity or repair probability.

Existing component	Proposed treatment
Duane discovery-rate and convergence telemetry	Retain reported findings at their stated telemetry scope. Validate any link to unknown residual flaws separately.
Stopping and substrate-ceiling rules	Treat as policies or hypotheses with stated assumptions. A low-yield pass does not establish exhaustion of all useful methods.

For a fixed prior $0 < \pi < 1$, the no-false-positive all-miss coverage relation remains

$$R = \frac{\pi(1-C)}{(1-\pi) + \pi(1-C)}, \quad C = \frac{\pi-R}{\pi(1-R)}.$$

This fixes the isolated inverse sign error and preserves the distinction between coverage and posterior risk. The map is not an invertible relation between arbitrary weighted aggregates.

The original definitions and extensions are in the [pinned appendix](#). The preceding review provides detailed line references. Any future implementation must change the documentation, forecasts, and dependent stopping rules together; a symbol-only substitution would conceal changed semantics.

6. Parameter uncertainty and limits of compression

The equation cannot turn unspecified inputs into knowledge. Record a prior's source and uncertainty. A default $R = 0.5$ is a symmetric assumption, not a universally conservative estimate. Seeded defects can help measure sensitivity to those seeds; their prevalence does not establish the natural prevalence of unknown defects.

Marginalize uncertain parameters using their **joint, correctly conditioned distribution**. Do not independently average every parameter and insert the averages. For example, after the evidence suppose two equally plausible parameter settings are

$$(B, s, b) = (0.8, 0.8, 0), \quad (0.2, 0.2, 0).$$

Both yield risk 0.16, so the joint average is 0.16. Separate means give $(0.5, 0.5, 0)$, incorrectly yielding 0.25. Conditioning removal on the initially flawed state instead gives marginal $s = 0.68$, which correctly recovers 0.16 at $B = 0.5$.

In general, with latent parameters θ ,

$$R_{\text{next}} = \mathbb{E}_{\theta}[(1-s_{\theta})B_{\theta} + b_{\theta}(1-B_{\theta}) | e, \mathcal{J}, a],$$

under the same declared policy. Before the evidence, the likelihoods also require the appropriate state-conditional parameter distribution.

A practical initial implementation can use a low-dimensional calibrated model with uncertainty intervals and sensitivity bounds. When the necessary inputs are unsupported, return **unestimated**, together with evidence and scope, instead of presenting a precise probability.

Scalar insufficiency has an elementary counterexample. Two targets each have probability $1/2$ of “some flaw.” In one population the only possible flaw is A; in the other it is B. A perfect negative test for A yields risk zero in the first and $1/2$ in the second. Equal initial risk does not imply equal future evidential response.

7. Applicability beyond a binary flaw

For several mutually exclusive states exhaustive within the declared model, use probabilities p_i , observation likelihoods ℓ_{ij} , and a transition matrix $T_{ij} = P(\text{new state } j | \text{old state } i, e, \mathcal{J}, a)$, with each row summing to one:

$$p'_j = \frac{\sum_i T_{ij} \ell_i p_i}{\sum_i \ell_i p_i}.$$

Require nonnegative state probabilities summing to one, nonnegative likelihoods, a positive denominator, and the same action-selection conditions as in the binary construction. The binary equation is its two-state projection. For competing claims about an unchanged world, no target-changing action means T is the identity. Scientific hypothesis revision changes the candidate description or state space and must be recorded explicitly.

For overlapping flaw classes, $\sum_k c_k R_k$ is expected additive loss when the consequences c_k are defined appropriately. It does not generally equal $P(\text{any flaw})$. Independence is unnecessary for linear expectation, but a union probability needs joint information; with only marginal risks,

$$\max_k R_k \leq P(\cup_k H_k) \leq \min(1, \sum_k R_k)$$

for the included classes. Interacting consequences require a richer loss function.

Scientific discovery can introduce a previously absent hypothesis or mechanism. A catch-all “other” category has no useful numerical prediction without an observation law. Record excluded possibilities, model misfit, and sensitivity to alternatives; do not invent a number claimed to quantify every unknown unknown.

The construction is substrate agnostic in its definitions: a human, an AI model, an instrument, or a team can produce evidence or take actions. This does not make their estimated parameters interchangeable. Shared data, correlated errors, complementary capabilities, selection, and cost determine whether a collective outperforms an individual. Neither “more agents always help” nor a universal single-agent capability ceiling follows from this equation.

8. Choosing actions, experiments, and stopping

After evidence has been collected, the expected reduction in this binary failure probability from an action is

$$\Delta_a = B - R_{\text{next}} = sB - b(1 - B).$$

It is positive precisely when $B > b/(s + b)$, provided $s + b > 0$. With $s = b = 0$, it is zero. This is a scoped risk comparison before adding costs or other consequences.

Experiment value is different. For a fixed hypothesis, expected information gain is

$$I(H; E | \mathcal{F}) = \mathcal{H}(R) - \mathbb{E}_E[\mathcal{H}(B_E)] \geq 0,$$

where $\mathcal{H}$ is binary entropy. A particular observation can increase uncertainty even though the expected information gain is nonnegative.

For decisions, define the loss of a final decision d in state h , $C(d, h)$. When an experiment j only supplies information, its one-step net decision value is

$$V(j) = \min_d \mathbb{E}[C(d, H) | \mathcal{F}] - \mathbb{E}_e[\min_d \mathbb{E}[C(d, H) | e, \mathcal{F}, j]] - c_j.$$

Costs must use compatible utility units, or be handled as explicit budget constraints. If a subsequent decision physically changes the target, evaluate its consequences through the transition model. If the experiment itself changes the target, model the actual action/observation order rather than treating it as passive evidence.

This is a one-step policy criterion, not a global optimum. Complementary experiments defeat naive greedy stopping: two independent fair bits individually reveal nothing about their XOR, but together determine it.

Where such dependencies matter, consider experiment bundles or limited lookahead, and label any heuristic.

The usable human rule is: **continue when an available investigation or sequence is worth its expected cost under the declared objectives; otherwise state the residual uncertainty and why work stopped.** Distinguish resolved acceptance conditions, a discovered counterexample, economical stopping, budget exhaustion, and unresolved blockage.

No general zero-risk limit is implied. With uninformative evidence and fixed action parameters, $R_{t+1} = b + (1 - s - b)R_t$. When $0 < s + b < 2$, its limit is $b / (s + b)$, since the deviation is multiplied by $1 - s - b$. At $s = b = 0$ nothing changes; at $s = b = 1$ the state generally alternates. These are conditional properties of that fixed process, not a law of scientific progress.

9. What would refute this proposal?

Separate a probability identity from a deployable model.

Claim	Evidence required	Failure or revision trigger
The compact equation follows from the definitions.	Written derivation and independent calculation.	A valid counterexample satisfying the definitions, or a flaw in the derivation.
The observation and action models describe a specified task family.	Prospective predictions and independently resolved outcomes.	Predeclared predictive discrepancies, drift, hidden selection, or omitted mechanisms.
Its numerical risks are useful probabilities.	Held-out calibration and forecast skill versus simple comparators.	Material calibration failure or no useful predictive advantage at stated precision.
Its policy improves research outcomes.	Matched-budget comparisons using external task outcomes.	No meaningful benefit, harmful changes, or unacceptable missed failures.
The richer model generalizes across STEM and substrates.	Transfer to new task families, domains, and participants.	Transfer failure; narrow the claimed scope.
CDSFL makes an original contribution.	A precise contribution statement and scoped literature comparison.	Prior work establishing the claimed contribution; revise attribution.

The broad core is intentionally flexible. For any posterior B and desired output y , choosing $s = 1 - y, b = y$ forces $R_{\text{next}} = y$. Therefore fitting those quantities after seeing outcomes can explain anything and tests nothing. The empirical claim must freeze or prospectively specify how parameters are obtained.

An unlikely event is not automatically a logical refutation of a stochastic model. Use predeclared predictive tests and practical discrepancy criteria. “No counterexample found” must not become “established true in all applications.” Model criticism and revision are central parts of applied Bayesian work, as discussed by [Gelman and Shalizi](#).

The accompanying [validation plan](#) provides a staged path that can begin before a distributed build-out.

9.1 Guard against conclusions hidden in assumptions

The supplied Hossenfelder transcript (archive: [work/math-review/Sabine_Maths_and_Circular_Reasoning_transcript.txt](#)), especially 03:05-03:52, describes the risk of a proof that assumes a condition containing the result it claims to establish. This review uses that methodological warning; it does not assess the transcript's claims about any particular current proof.

For each substantive claim, record its exact conclusion, dependencies, and the support for each dependency. Expand important lemmas far enough to ask whether an assumption simply restates the conclusion in different language. A dependency diagram without a visible cycle is insufficient if the conclusion is concealed inside a premise.

For this proposal the dependency audit is:

Claim	Permitted starting point	What would make the argument circular or unsupported
The update identity is correct.	Probability axioms, explicit conditional-event definitions, and the stated selection condition.	Treating the identity as proof that supplied empirical inputs are calibrated.
A test could expose the specified failure.	Independently resolved reference cases, a justified analytical operating characteristic, or clearly labelled uncertainty.	Inferring high test sensitivity merely because the current target passed or because the reviewer agreed with itself.
A repair reduces real failure risk.	Scoped before/after outcomes under the relevant selection policy.	Calling a repair successful because the model assigned it a high success parameter.
CDSFL improves research.	Prospective external outcomes against fair comparators.	Defining success as a decrease in the very score the procedure is designed to decrease.
A theoretical result is proved.	Independent premises and valid deduction over the stated domain.	Assuming an unproved regularity, existence, solvability, or error bound that already contains the desired conclusion.

Definitions of s, b, ℓ_1, ℓ_0 do not establish their numerical values. The mathematical claim is only the conditional update. No assumption that CDSFL works appears in its derivation, and no conclusion that CDSFL works follows from that derivation alone.

Constructed outcome trees check the calculation within supplied worlds. They cannot establish that those worlds describe actual research. A simulation designed to follow the equation is therefore an implementation test, never the sole empirical validation of the model.

9.2 Connect severe tests to the original coverage model

One operational requirement of a severe test is that it would seldom **accept** a target containing the specified failure. Let

$$\beta = P(\text{pass} | H, \mathcal{F}), \quad \gamma = P(\text{pass} | \neg H, \mathcal{F}).$$

Pass must be a predefined observed outcome on an unchanged target. It is distinct from timeout, abstention, or an unresolved investigation. Suppose external calibration or a justified analytical argument supports $\beta \leq \beta_{\max}$ and $\gamma \geq \gamma_{\min} > 0$. For $0 < R < 1$,

$$R_{\text{pass}} = \frac{R\beta}{R\beta + (1-R)\gamma} \leq \frac{R\beta_{\max}}{R\beta_{\max} + (1-R)\gamma_{\min}}.$$

The proof is short: posterior odds equal prior odds multiplied by β / γ , which is at most $\beta_{\max} / \gamma_{\min}$; conversion from odds to probability preserves order.

For a **completed binary protocol**, pass and failure detection are exhaustive. Then original coverage is $C = 1 - \beta$, and if the false-alarm probability is α , $\gamma = 1 - \alpha$. History-conditional coverage retains

$C = 1 - \prod_i (1 - q_i)$. This explicitly preserves the original insight: survival is informative according to the examination's ability to expose a relevant error.

As an illustration only, bounds $\beta \leq 0.05$, $\gamma \geq 0.95$, and prior $R = 0.5$ imply $R_{\text{pass}} \leq 0.05$. These are stipulated numbers, not estimates obtained for CDSFL.

A hidden partition assumption would invalidate that calculation. Suppose flawed targets have probabilities (pass, fail, abstain) = (0.05, 0.95, 0), while clean targets have (0.05, 0.05, 0.90). The miss and false-alarm rates are both 0.05, but $\gamma = 0.05$, not 0.95. A pass leaves prior 0.5 unchanged. This case was found during the additional challenge and is included in the checks.

Calibration evidence must apply to the actual target family, test-selection process, and available information. Statistical confidence bounds are not infallible assumptions: propagate their uncertainty and joint coverage. Two separate 95% bounds do not automatically supply a joint 95% guarantee. Average detection across a benchmark also does not establish the same power for every failure subtype. A universal claim requires suitable control across the stated alternatives, or a narrower conclusion.

This operational connection is not a claim to encode every philosophical meaning of severity. It supplies a concrete quantity that experiments can estimate and contradict. The original paper already demands serious attempted refutation, distinguishes corroboration from proof, and permits replacing its equation when another predicts outcomes better. See [P-Pass](#) and [Invitation to Falsify](#).

10. Applying the revised framework to this revision

The requested self-check is rational as an examination of consistency and evidence. It is not independent validation of the framework's own assumptions.

Review claim	Evidence and action recorded	Present status
The earlier repair semantics fail in the elementary example.	Explicit positive/negative outcome tree; old and revised calculations compared.	Counterexample established under the stated repair policy.
The new core has a valid probability interpretation.	Bayes plus total probability; separate derivation; exact outcome enumeration.	Established within the definitions and conditioning requirements.
Valid original coverage and clean-review results survive.	Algebraic reductions and exact regression cases.	Established within their original valid assumptions.
The proposal handles the identified selection and dependence failures.	Private-selection, duplicate-evidence, state-insufficiency, and joint-uncertainty counterexamples; explicit conditions added.	Those failures addressed; no claim of exhaustive discovery.
The severe-test bound remains valid with unresolved outputs.	An initially binary-only formulation failed on an abstention counterexample. Replaced it with explicit pass probabilities under flawed and clean states.	Corrected conditional bound derived and checked; empirical rates still unestablished.
The inputs are calibrated for real CDSFL work.	No prospective calibration dataset analysed in this revision.	Unestablished.
The revision improves actual STEM research or collective performance.	No matched-budget deployment experiment performed.	Unestablished.

Review claim	Evidence and action recorded	Present status
The equation is historically novel.	It is derived from standard probability identities. No exhaustive priority review performed.	No new-theorem claim; application-level contribution remains to be assessed.

The old model's useful principles were applied: state claims and constraints, search for disproof, preserve evidence, revise after counterexamples, and distinguish external anchoring from internal coherence. Applying the revised version added explicit target versions, observed branches, action semantics, dependent evidence, and unestimated quantities.

Three delegated passes supplied derivation, adversarial challenge, and validation design. These were instances of the same model and shared context or assumptions. They are separate checks, not independent empirical samples. Reviewer agreement was not multiplied into a confidence estimate.

The reference suite passed **5,600 conditional outcome-tree cases, 650 impossible-observation rejections, 2,000 branch-policy expectations, 125 sequential-repair cases, and 125 reduction/expectation cases**. Eight deliberately incorrect alternatives were rejected. These are constructed exact-arithmetic checks, not 8,500 independent scientific experiments. The general derivation supplies the proof; the finite cases check consequences and implementation.

Files: reference arithmetic (archive: outputs/CDSFL_revised_core.py), reproducible checks (archive: outputs/CDSFL_revised_model_checks.py), and recorded results (archive: outputs/CDSFL_revised_model_check_results.json). Separate supporting scripts also passed overlapping checks; their counts are not added to the main suite. No formal proof assistant or real-world benchmark reproduction was used.

Revision 1.1 adds severe-test checks (archive: outputs/CDSFL_severe_testing_checks.py) and their results (archive: outputs/CDSFL_severe_testing_check_results.json): 450 constructed bound cases passed, and three misleading alternatives were rejected, including the newly identified abstention error. These test the stated bound and its limits; they do not estimate real test sensitivity.

No numerical “probability that this model is sound” is reported. Its requisite priors and reviewer error likelihoods have not been calibrated. This is the revised discipline applied to our own claims.

11. Keep only components that earn their place

The mandatory core contains the scoped event/history, actual-evidence update, and an action transition only when the target can change. Every further component needs a stated job:

Component	Job	When to use it
Coverage/severe-test bound	Explain how a passed test bears on a specified failure.	When its operating characteristics can be justified or estimated.
Joint parameter uncertainty	Prevent false precision and incorrect averaging.	Whenever uncertain inputs materially affect the conclusion.
Multiple states or flaw classes	Represent distinctions that alter predictions or consequences.	When the binary projection loses decision-relevant information.
Expected loss and experiment value	Decide between actions or further investigation under resource constraints.	When making that decision, with stated objectives and costs.
Information gain	Compare learning about a fixed hypothesis.	When learning is an explicit objective; it is not a substitute for practical outcomes.

Component	Job	When to use it
Fixed-process limit	Check or refute a claimed convergence/floor property.	Only for a process satisfying the fixed-rate assumptions.
Using or other richer dependence model	Predict dependencies missed by a simpler calibrated model.	After a relevant failure or comparative result justifies the complexity.

No optional equation is a compulsory implementation layer. An extension that adds neither a necessary correctness condition nor demonstrated explanatory, predictive, or decision value should be omitted from deployment.

12. Recommended next work, when the existing build is ready

1. **Adopt the mathematical specification as a candidate revision.** Preserve the old equations and objections as versioned history. Have a probability/statistics reviewer and a domain researcher challenge the definitions and the worked cases.
2. **Create a small compatibility layer.** Store evidence updates, target-changing actions, provenance, and versioned events separately. Keep unestimated parameters visible. Do not silently compare old and new risk values as the same metric.
3. **Run a cheap diagnostic pilot, then lock one task family.** Separate observing frozen targets from repairing cloned targets. Use independently resolved outcomes and simple baseline models.
4. **Evaluate predictive value before complexity.** Add richer dependence, latent states, and adaptive policies only where they improve held-out performance or address demonstrated failure.
5. **Test transfer and collective benefit.** Only then claim broader STEM usefulness or a measured advantage from coordinating more capable and less capable participants.

A compact public statement could be:

CDSFL organizes scientific enquiry around explicit claims, discriminating evidence, auditable revisions, and testable stopping decisions. Its tests must be capable of exposing the failures they claim to exclude. Its mathematical core updates belief about a defined failure and predicts how an action changes that failure. Its reliability in any application is measured against external outcomes within a stated scope.

This preserves a meaningful claim about scientific enquiry while leaving the system open to refutation. The sound conditional identity can survive changes in the machinery around it. The estimated probabilities, scientific scope, and demonstrated usefulness must remain revisable.

CDSFL lineage and decay: clarification of the revision

10 September 2026 (London). Companion to the version 1.1 mathematical proposal. This records a clarification requested by George Jackson; the core specification and equations are unchanged.

Conclusion: the proposal is a revision of CDSFL's model with explicit mathematical lineage. It is not a wholesale replacement. The valid coverage and observation mathematics survives exactly. Discovery-rate models can remain central empirical companions. The repair interpretation and novelty-validity coupling require replacement, and some universal interpretations of decay require correction.

Human explanation and standalone status

The founder clarified the cumulative wording to:

Capable repeated examinations progressively reduce the opportunity for an existing error to remain undetected. Their returns can diminish. Repairs can remove errors and introduce others. Discovery curves help us understand whether further investigation is productive.

The first sentence concerns an unchanged target and a consistently defined error. Each further examination must retain some ability to detect an error that earlier examinations could have missed; duplication alone need not add coverage. If the target or criterion changes, the revised state must be assessed explicitly.

This is an independently assessable account of scientific enquiry, with CDSFL as a possible implementation and testing vehicle. The revision makes that prospect better specified and easier to challenge: it preserves the valid original dynamics, removes identified semantic errors, and separates formal derivation from empirical adequacy. It does not establish historical novelty, usefulness across all STEM, or experimental success by definition.

1. The collapsed form recovers the original recursion exactly

The revised core is

$$R_{next} = \frac{(1-s)LR + b(1-R)}{LR + 1 - R}.$$

For an unchanged target, set $s = b = 0$. After a clean review with sensitivity p and zero false positives, $L = 1 - p$. Substitution gives

$$R_{n+1} = \frac{R_n(1-p)}{1 - pR_n},$$

the original valid CDSFL residual-risk recursion.

Starting from $R_0 = \pi$, repeated clean reviews give

$$R_n = \frac{\pi(1-p)^n}{1 - \pi + \pi(1-p)^n}.$$

The elementary proof is in odds: each clean result multiplies the previous odds by $1 - p$, so after n results the multiplier is $(1 - p)^n$. Conversion back to probability yields the displayed expression.

Assumptions: the same scoped failure and unchanged target; actual negative results; constant sensitivity conditional on preceding misses; zero false positives; a supported observation history. Independent constant-sensitivity reviews are a sufficient special case. Ordinary illustrations use $0 < \pi < 1$, $0 < p < 1$; impossible boundary observations remain undefined.

2. The original decay curve survives

Quantity	Expression	Meaning
Conditional miss probability	$m_n = (1 - p)^n$	Probability every review misses, if the specified flaw exists.
Coverage	$C_n = 1 - (1 - p)^n$	Probability at least one review would detect that flaw.
Next increment in coverage	$C_{n+1} - C_n = p(1 - p)^n$	The diminishing additional detection opportunity.
Posterior odds	$R_n / (1 - R_n) = [\pi / (1 - \pi)] (1 - p)^n$	The same geometric decay expressed as evidential odds.

The factor driving the original curve is literally retained. It was not replaced by an unrelated law.

Coverage and residual risk report different quantities. With a fixed prior,

$$R_n = \frac{\pi(1 - C_n)}{1 - \pi + \pi(1 - C_n)}.$$

This is a change of reported quantity, not a claim that risk and coverage are numerically identical. In particular, high coverage is conditional detection ability, not automatically the probability that a target is correct.

The posterior probability R is not itself an exponential. Its absolute decreases also need not diminish from the first review onward. For example, prior 0.99 and sensitivity 0.5 produce initial decreases about 0.009802, 0.019033, and 0.035931. This corrects an overgeneralization about R ; it does not refute geometric miss-probability decay or diminishing coverage increments.

The original equations are in [PAPER §2.1](#) and the [appendix's recursive derivation](#).

3. Expanded-form heritage

The relationships should be named accurately:

Original element	Relationship to the proposal
$C(n)$, single-class coverage	Preserved exactly under its valid assumptions.
F , weighted multi-class coverage	Retained as a defined coverage report; its inputs require justified conditional detection probabilities or calibrated approximations.
R , residual risk after clean evidence	An exact special case of the revised observation update.
Human/machine coverage	Retained through conditional incremental detection; statistical independence is not inferred merely from different sources.

Original element	Relationship to the proposal
Actual positive results and false alarms	The observation model is extended to include them explicitly.
Repair and reinjection	Replaced by a derived state transition separating what was learned from what was changed.
Literature novelty	Retained as a separate scientific/reporting dimension, without automatically suppressing valid replication evidence.
Discovery-rate and convergence curves	Retained as testable empirical models and decision inputs, with their meanings distinguished from residual risk.

Some links are exact special cases, some are transformations between quantities, and some are empirical companion models. Claiming that every original formula is a parameter setting of a single collapsed scalar would misstate the lineage.

4. Duane decay is preserved at its proper scope

The original appendix uses

$$\lambda(t) = \frac{\beta}{\eta} (t / \eta)^{\beta - 1}.$$

For $\eta > 0$ and $0 < \beta < 1$, this describes a declining power-law discovery intensity. It remains available for modelling verified findings, comparing observed trajectories, predicting near-term yield, and informing resource decisions. It is not displaced by the revised risk update, which answers a different question.

The power-law NHPP has an established reliability-modelling basis; see [NIST's Duane model](#). Its application to CDSFL finding counts is an empirical modelling choice. The [project appendix §7.1](#) reports a better AICc fit than geometric decay in 17 of 18 runs. This clarification preserves that reported result at its stated scope; it does not reproduce or independently confirm the fit.

The required corrections concern inference from shape:

- A declining finding rate can reflect depletion, but also reduced sensitivity, narrowing attention, or loss of useful context.
- A constant finding rate can reflect continuing genuine discoveries; it is not by itself proof of churn.
- A later increase can reflect a better method or newly opened line of investigation, not necessarily a deterioration in reasoning.
- The standard-error relation proportional to $1 / \sqrt{n}$ for suitable repeated measurements does not establish a universal law for the rate of discovering new defects.

The project's separate verification, novelty, rediscovery, and efficiency measures are useful precisely because one curve does not identify its own cause.

A count-rate model can supply evidence to the revised core if its observation likelihoods are justified under the competing failure states. A fitted curve alone does not supply those two likelihoods. This is a possible calibrated connection, not permission to equate falling counts with falling risk.

5. What the correction establishes

The original model had real merit: its coverage calculation, Bayesian clean-review interpretation, explicit attention to test capability and dependence, intelligible decay behaviour, and commitment to external falsification provide a substantive foundation.

Parts survive unchanged; other parts are corrected or replaced. That is successful revision of a research framework, not confirmation of every earlier claim. Mathematical preservation does not itself corroborate the empirical claim that a particular fitted decay law describes all STEM enquiry.

The appropriate description is: **CDSFL's existing mathematical foundation, revised to separate evidence, intervention, and discovery dynamics, while preserving valid decay relationships and their human-comprehensible interpretation.**

Work record for this clarification

- Re-read the original coverage, structured model, recursive risk, Duane, and discovery-efficiency passages in the frozen repository snapshot.
- Compared them with version 1.1, especially its retained-lineage section.
- Requested a bounded independent lineage derivation; it confirmed the recursion, closed form, odds decay, and diminishing coverage increment using exact arithmetic.
- Checked NIST's primary description of the Duane/NHPP power-law model.
- No empirical benchmark was rerun, and no upstream or core-specification files were edited. The exact identities and the scope of reported empirical findings remain distinct.

CDSFL revised model: prospective validation and refutation plan

Version 1.1 proposal - 10 September 2026 (London). Companion to the [revised mathematical specification](#). The structural mathematical checks have passed. No empirical calibration or deployment experiment was performed in this revision exercise.

This plan is designed to begin on a small scale after the current build work permits it. A successful first task family would support a bounded usefulness claim. Broader STEM and substrate generality require additional evidence.

1. Make the empirical claim capable of failing

The core equation is a probability identity when its quantities have the stipulated meanings. An operational model also claims that its **chosen states, estimated observation likelihoods, action transitions, and decision policy** describe a real task family. Those commitments make prospective predictions.

Before each evaluation, freeze:

- The target family and sampling procedure.
- The failure event, acceptance criterion, and independently available outcome check.
- The model version, parameter-estimation procedure, and allowed adaptation.
- The comparison models and policies.
- The practical improvement sought, tolerated regressions, sample-size rationale, and analysis plan.

Do not choose removal or introduction parameters after seeing the answer. The general equation can fit any desired result if those are left unconstrained.

Maintain separate claims for mathematical correctness, predictive adequacy, useful decisions, transfer, and originality. A stochastic forecast is not logically refuted merely because a low-probability event occurs. A predeclared predictive test can reject adequacy at its stated error rate; a practical discrepancy can also justify changing a model without pretending it was a deductive contradiction.

Applied Bayesian practice includes model construction, checking, comparison, and revision, not only posterior computation. See [Gelman and colleagues, Bayesian Workflow](#). The CDSFL-specific programme below is a proposal, not a result from that paper.

2. Define events that can actually be resolved

An initial event might be:

This numerical routine violates property P on input distribution D, within the stated tolerance and evaluation procedure.

“Some unknown flaw exists somewhere” usually cannot receive a complete independent label. If the oracle checks only a defined class or finite domain, restrict the claim to that scope.

Record before observing the result:

Record	Minimum content
Target	Identifier, version/hash, source family, relevant assumptions.

Record	Minimum content
Event	Exact failure/acceptance criterion and scope.
Prediction	Prior or posterior forecast, parameter provenance, and uncertainty; unestimated where necessary.
Evidence	Identifiers, collection/selection procedure, available history, dependencies, and target version examined.
Action	Chosen action and policy, relevant selection information, resulting target version.
Outcome	Independent resolution, its scope, ambiguous/missing labels, and any oracle limitations.
Resources	Actual cost, elapsed time, and why work stopped.

Keep a source artifact and its mutations in the same development or held-out partition. Many mutations of one target do not supply many independent task families.

A changed proof or design gets a new version. A revised claim does not retroactively become the original claim. A timeout is unresolved rather than a clean review. Missing outcomes must remain visible; otherwise successful-to-verify tasks can dominate the apparent performance.

3. Calibrate observation, action, and verification separately

Observation cohort: what can the review detect?

Use frozen targets whose scoped positive and negative labels are independently established. Run the review without modifying the target. Estimate

$$q = P(\text{positive result} \mid \text{flaw, context}), \quad f = P(\text{positive result} \mid \text{no flaw, context}).$$

Include both valid and invalid artifacts. A collection containing only planted defects cannot estimate false-positive rate. Seeded defects support claims about detection of those defect types; seeded prevalence does not establish natural-world priors.

For a genuinely comparable Bernoulli cohort, uncertainty about a rate can be represented with a declared binomial model and appropriate interval or prior. The denominator is independently labelled opportunities in the relevant cohort, not an arbitrary count of generated assertions or tests chosen after the result.

Reviewers' self-reported findings alone generally do not identify the natural flaw rate, sensitivity, and false-positive rate separately. Use independently resolved labels or make the additional identifying assumptions explicit.

Action cohort: what do repairs change?

Use cloned target snapshots and a declared intervention policy. Include initially flawed and initially clean targets to estimate

$$s = P(\text{post clean} \mid \text{pre flawed, evidence, action}), \quad b = P(\text{post flawed} \mid \text{pre clean, evidence, action}).$$

“Clean” here means passing the scoped independently resolved criterion. Removing one known flaw is not sufficient if the event is “at least one failure in this class remains.”

Retain a no-action comparison where practical. Randomized action assignment within relevant strata makes policy comparisons easier to interpret, where feasible. If actions are selected after positive results, estimate their effects in that population. An unconditional repair rate need not apply.

Initially clean targets selected for repair can be rare false-positive cases. Deliberately enriching those cases can help measure damage, but natural expected policy outcomes still require the correct population and selection weights. Do not mistake the enriched mixture for deployment prevalence.

Verification cohort: is the changed target acceptable?

Resolve post-action outcomes using a check that is independent of the repair's optimization target as far as the scoped task permits. Retain the dependency record when independence is incomplete. Passing the exact examples a repair was designed to pass cannot by itself serve as a broad correctness oracle.

Reuse of the same checker on a changed target can still produce evidence. The required question is what that result implies under the actual development and selection process.

4. A progression suited to a sole founder

Stage	Work	What success supports
0. Structural mathematics	Exact derivation, outcome enumeration, boundaries, and deliberately broken alternatives. Completed for this revision.	Internal correctness of the supplied finite-state model and reference arithmetic.
1. Diagnostic pilot	About 12 distinct tasks with independently resolvable outcomes, covering true/false positives, missed flaws, failed repairs, harmful changes, and evidence reuse.	Logging and oracle feasibility, runtime/cost estimates, and protocol corrections. Not certification.
2. First locked family	Mathematical or numerical artifacts with independent symbolic, exhaustive, or reference-computation outcomes; development fit followed by untouched evaluation.	Predictive and decision usefulness within that defined family.
3. Mechanistic inference	Known simulated mechanisms, noise, correlated observations, new data, and a true mechanism deliberately omitted from the initial candidate set.	Controlled transfer and detection of specified model misspecification.
4. Real STEM transfer	A reproducible scientific-computing or experimental-replication task with withheld observations and a domain-appropriate reference procedure.	Evidence about that real domain, including limits of its outcome labels.
5. Broader transfer	Another domain; different models, instruments, and eventually human participation where such claims are intended.	A progressively broader, still bounded generality claim.

The pilot size is a design suggestion, not a power calculation. Pilot results remain development data. Choose the locked evaluation size from a prespecified practical tolerance, effect size, and uncertainty analysis informed by the pilot. If adequate precision is unaffordable, retain wider uncertainty and narrow the claim.

Some theoretical scientific claims remain unresolved. In that case evaluate observable predictions and decision consequences; do not relabel consensus as known truth.

These early stages do not require building a global distributed platform. They establish whether the mathematical instrument measures anything useful before increasing its operational scale.

5. Compare forecast quality and decision quality separately

Use three inexpensive forecast comparators:

1. A base-rate-only prediction.
2. A simple calibrated observation/action model with few parameters.
3. The proposed richer model, adding dependence or latent structure only where specified.

Use the former CDSFL score as a labelled heuristic comparator unless its probability interpretation has been established. Evaluate forecasts using the same unseen evidence; evaluate complete research policies separately under matched resource budgets.

For binary outcomes, a suitable initial primary metric is Brier loss,

$$\frac{1}{n} \sum_{i=1}^n (p_i - y_i)^2,$$

where p_i is the frozen probability of the event and y_i its independently resolved zero/one outcome. Add log loss and reliability summaries as diagnostics. Proper scoring rules assess probabilistic forecasts, while calibration and useful discrimination need separate attention. See [Gneiting and Raftery \(2007\)](#).

A constant base rate can be calibrated yet uninformative. Report forecast skill relative to baselines, predeclared subgroups, and task-family uncertainty. Aggregate performance can hide overconfidence on difficult or unfamiliar targets. For log loss, address exact zero/one forecasts prospectively; do not quietly clip failed predictions after outcomes become known.

For decision policies, measure:

- Independently resolved task outcomes and missed unacceptable failures.
- Unnecessary actions and newly introduced failures.
- Total resource cost and elapsed time.
- Useful replication, discovery, and other declared scientific objectives separately where they are not captured by one loss measure.

A lower internally computed risk is not the success endpoint. Prefer the simpler model if added complexity has no demonstrated practical benefit.

Predeclare a practically meaningful difference and tolerated harms. Use paired comparisons where the same independent task families can support them, with uncertainty at the family level. Treat a wide interval as inconclusive, rather than evidence of equality.

Small samples constrain claims. With independent identically distributed Bernoulli trials and zero observed failures, a one-sided 95% upper confidence bound is $1 - 0.05^{1/n}$, about 9.5% at $n = 30$ and 3.0% at $n = 100$. This follows by solving $(1 - p)^n = 0.05$; it does not transfer to correlated review passes or a different deployment distribution. See [NIST on exact binomial limits](#).

6. Stress the assumptions that can make the system overconfident

Dependence: repeat exactly the same evidence, use reviews that share the first review's conclusions, and collect genuinely new replication data. The first should add no information once recorded; the last may add evidence even when the result is already published. Different model names do not establish independence.

Selection: compare fixed and adaptive policies prospectively. Record the information available to the selector. Log available alternatives and selection probabilities if later counterfactual policy evaluation is intended. A deterministic policy provides no direct outcome data about actions it never chooses.

State insufficiency: use targets with equal initial scalar risk but different hidden flaw types. A test aimed at only one type should have different implications. This challenges any claim that one number alone determines the next update.

Parameter dependence: compare joint uncertainty propagation with separate parameter averages. Include cases where harder targets also have poorer repair success, and where positive-result selection changes the repair population.

Unknown alternatives: deliberately omit the true mechanism in controlled experiments. A model can be confidently wrong within an incomplete candidate set. Maintain discrepancy checks and a route to revise that set; a generic “other” label without predictions does not solve the problem.

Complementary experiments: include a task where two individually uninformative measurements are jointly useful. This tests stopping and allocation policies rather than only the probability arithmetic.

Changed targets: test whether evidence about the original artifact is improperly carried over as fresh confirmation of a repair, and whether introduced failures in other classes are missed by a narrow repair-success measure.

Changing domain, model version, tools, target family, or selection policy triggers an applicability review. Recalibration may be appropriate, but transfer failure remains in the record.

7. Predeclare how to stop and what to publish

For the first empirical study, use a fixed sample and analysis plan. If repeatedly inspecting results to decide when to stop, use a valid time-uniform method under its assumptions or reserve a fresh final evaluation. Do not repeatedly tune against the same holdout until the claim passes. Time-uniform confidence sequences provide one established approach; see [Howard and colleagues](#).

A tool score called an “e-value” is not thereby an e-value. Any sequential evidence gate requires the actual statistical construction and null-expectation guarantees.

For operational work, record one of the following: acceptance conditions resolved; counterexample found; economical stopping with residual uncertainty; budget exhausted; blocked/unresolved. Budget exhaustion does not mathematically prove or disprove the target claim. It also should not cause a numerical prior inflation without an observation model linking that event to the failure risk.

Publish outcomes, failures, uncertainties, task exclusions, missing labels, costs, and the frozen model version. Distinguish a failed scientific hypothesis from a failed implementation, and both from a study too imprecise to decide. This record is essential when later improvements might otherwise erase the evidence of earlier limitations.

8. Operational severe testing and non-circular evidence

Before calling a successful test reassuring, ask how likely its observed pass would be under the specified failure and under its absence. Estimate those two pass probabilities directly when timeout, abstention, or unresolved outcomes are possible. A low false-alarm rate does not imply a high clean-pass rate if most clean cases produce no answer.

The specification derives a residual-risk upper bound from a justified upper bound on flawed-target acceptance and a lower bound on clean-target acceptance. When estimated from data, those bounds need uncertainty treatment and joint coverage. Report whether detection ability is averaged over a reference population or controlled across all alternatives claimed; these support different conclusions.

The new abstention counterexample is a required structural case. Also challenge the test with alternatives resembling the errors the actual research task might contain. A method that detects conspicuous planted faults can still be weak against the relevant subtle failure.

Audit each theoretical conclusion for hidden assumptions that already contain it, including differently named regularity or error bounds. Every important dependency needs independent support, a valid derivation from earlier premises, or an explicit unresolved/conditional label. Self-generated numerical confidence and internally consistent derivations cannot supply missing empirical support.

Experiments can demonstrate useful outcomes within a stated domain and refute precise predictions. They do not establish universal truth from a finite run of successes. Exact mathematical claims need proofs from adequate independent premises or valid counterexamples. The two kinds of evidence remain complementary.

9. Current self-audit and next decision

Claim	Current status
Collapsed equation follows from the stated model.	Established by the written derivation; reference calculations passed.
Main earlier repair and novelty counterexamples are addressed.	Established for the stated constructed cases.
Observation and action probabilities are calibrated for CDSFL deployment.	Unestablished.
Revised policy improves STEM research outcomes at a useful cost.	Unestablished.
Benefits transfer across domains and substrates.	Unestablished.
A distinct scholarly contribution has been demonstrated.	Not established by this review; requires precise comparison and evidence.

The next empirical decision is whether a small, independently resolvable task family can support meaningful estimates of observation and action behaviour. If it cannot, revise the event definition or measurement plan before increasing model complexity.

The revised framework is applied to this review through explicit claims, evidence provenance, recorded corrections, and honest unestimated quantities. No self-assigned probability of mathematical or scientific soundness is manufactured from agreement among AI reviewers.

CDSFL mathematical review - 9 September 2026

Scope: the mathematical claims in Constraint Engineering at commit 9e3dfe9d68b428acb98e65c0ad3800ec571a4da4. This is an assessment of the written model, its probabilistic interpretation, and the evidence described in the repository. It is not a full source-code audit or an empirical reproduction of the benchmark programme. No project files were changed.

Verdict: the independent coverage identity and the Bayesian update after non-detection are sound under explicit assumptions. The complete current model cannot receive an unqualified endorsement as a sound general account of scientific enquiry. In particular, the repair extension conflates conditioning on evidence with changing the object being assessed, and the literature-novelty extension conflates originality with evidential contribution. These are questions about the mathematical model's meaning, separate from harness faults. There are also smaller, reproducible algebraic and scope errors. The possibility of a valuable scientific contribution remains open.

This review used separate algebraic, probabilistic, and evidential review passes. The conclusions rest on the definitions, derivations, and counterexamples below, rather than on a vote among reviewers. Numerical checks were independently executed using exact rational arithmetic. The companion Python script (archive: outputs/CDSFL_math_counterexamples.py) uses only the standard library and reproduces the main examples.

1. The valid core and its scientific meaning

Let H mean a specified flaw exists in an unchanged target. Let D_i mean review i detects that flaw. If

$$q_i = P(D_i \mid H, \text{all earlier reviews missed}),$$

then the chain rule gives

$$P(\text{all reviews miss} \mid H) = \prod_i (1 - q_i).$$

Therefore $C = 1 - \prod_i (1 - q_i)$ is the probability this review procedure would detect the flaw if it existed. With independent reviews of constant sensitivity p , this becomes $C(n) = 1 - (1 - p)^n$.

This is valid probability mathematics. It formalises a meaningful aspect of enquiry: failing to find an error is informative to the extent that the examination could have exposed it. Independence is sufficient for substituting constant marginal probabilities. More generally, correctly specified conditional probabilities can handle adaptive reviews. Fixed diversity discounts are not automatically those probabilities.

$C(n)$ is a property of detection coverage conditional on an existing flaw. It is not, by itself, the posterior probability that a claim is true. The project's separate risk equation addresses that distinction.

Let $R = P(H)$, $q = P(D \mid H)$, and suppose there are no false positive detections. After a negative result,

$$R_{\text{minus}} = P(H \mid \text{no detection}) = R(1 - q)/(1 - qR).$$

This is exactly Bayes' theorem. In odds form,

$$R_{\text{minus}}/(1 - R_{\text{minus}}) = (1 - q)R/(1 - R).$$

Sequential updating and the batch posterior consequently agree. The prior is retained in the running posterior; it does not cease to matter.

If false positives occur with probability f , the denominator becomes $R(1 - q) + (1 - R)(1 - f)$. The simple recursion therefore needs either the zero-false-positive assumption or an appropriate observation model.

Sources: [PAPER §2.1](#), [appendix residual risk](#), [recursive derivation](#). The appendix's intellectual lineage itself acknowledges that this is an application of Bayes' theorem. See also [NIST's account of Bayesian reliability](#).

2. The principal modelling defect: non-detection is not repair

The three-phase extension defines:

- $R_{det} = R_{old}(1 - q)/(1 - qR_{old})$
- $R_{base} = \sigma R_{det} + (1 - \sigma)R_{old}$
- $R_{new} = (1 - \nu)R_{base} + \nu$

Here σ is described as the probability/efficacy of successfully resolving the detected flaw, and ν as reinjection. The first expression, however, is the posterior conditional on detecting nothing. It cannot simply become the probability after detecting and repairing a flaw because it is renamed “Detection.”

A complete elementary experiment makes the distinction visible. Suppose:

- Before review, a target has probability 1/2 of containing one flaw.
- The review detects an existing flaw with probability 4/5.
- There are no false positives.
- Every detected flaw is repaired perfectly.
- Repairs introduce no other flaws.

Observed outcome	Probability of outcome	Flaw probability after observation	Flaw probability after repair
Detection	2/5	1	0
No detection	3/5	1/6	1/6

The expected residual flaw probability after the cycle is therefore

$(2/5) \times 0 + (3/5) \times (1/6) = 1/10.$

The CDSFL three-phase recurrence returns 1/6. That is correct conditional on the negative-result branch, but it is neither the expected residual across the cycle nor the residual after a confirmed perfect repair.

A second boundary test is even simpler. With a confirmed existing flaw, $R=1$, $q=4/5$, $\sigma=1$, $\nu=0$, the written recurrence returns $R_{new}=1$. Under its stated perfect-repair meaning, the probability that the target flaw remains should be zero. This example does not use the excluded $R=q=1$ boundary.

This is a semantic problem in the probability model, not merely a numerical approximation or an uncalibrated parameter. The convex interpolation is a well-defined bounded function and could be assigned another interpretation, such as a scoring heuristic or an explicitly specified latent mixture. Its stated repair interpretation has not been derived.

A possible repair to the formal model would separate:

1. Observation: update the probability of the old target being flawed, conditional on the actual positive or negative result.
2. Intervention: model whether a repair changes that target's state.
3. New errors: model flaws introduced into the new target version.
4. Subsequent verification: collect evidence about the changed target.

For this elementary single-flaw experiment, the ex-ante post-repair probability is $R(1 - q \sigma)$. If an independent new flaw is then introduced with probability ν , it is $\nu + (1 - \nu)R(1 - q \sigma)$. This is an illustration under specified assumptions, not a proposed universal replacement equation.

Sources: [appendix three-phase extension](#), [operational directive §3](#). The directive substitutes a tool-derived solution score S_k for sigma and splits reinjection into two terms; that does not by itself correct the observation/intervention distinction.

3. Scientific novelty and evidential strength must remain distinct

Stage 6 specifies

$$\text{eta_combined} = \text{eta_int} [1 - \text{c_ext}(1 - \text{nu_lit})]$$

and uses $q = \text{eta_combined} d p$ in the residual-risk update. Here nu_lit denotes literature novelty, to distinguish it from reinjection nu .

At $\text{eta_int}=1$, $\text{c_ext}=1$, $\text{nu_lit}=0$, the effective detection probability is forced to zero. A fully known result then contributes no reduction in the stated residual risk.

This is appropriate only if the quantity being measured is something such as credit for original discovery. It is not generally correct as a rule for scientific corroboration. Independently replicating a published experiment, or applying an established diagnostic to a newly examined target, can supply strong fresh evidence while having no publication novelty. Established mathematics can expose a previously unnoticed flaw in a new derivation without itself becoming a new theorem.

Internal redundancy can matter to conditional information gain. Publication novelty answers a different question. Keeping both dimensions in reports does not remove their mathematical coupling inside q .

At prior $R=1/2$ and diagnostic sensitivity $p=4/5$ with $d=1$, a clean, informative established test gives posterior $1/6$. Stage 6's zeroed q leaves it at $1/2$.

For a model intended to describe scientific enquiry, replication and the productive application of established knowledge must remain possible sources of corroboration. Literature novelty can be tracked independently or used for discovery-priority decisions without automatically changing the likelihood of detecting a flaw.

Source: [appendix Stage 6 definition and boundary cases](#). The critique concerns the written extension; it is not an assertion that every live harness path currently applies it.

4. Further mathematical findings

Claim or mechanism	Assessment	Reproducible example or condition
Batch posterior equals recursive posterior	Valid under the observation assumptions	The exact-arithmetic script verifies a heterogeneous sequence, in addition to the algebraic derivation.
Break-even $\text{nu_star} = \sigma R q / [1 - q R (1 - \sigma)]$	Correct for the stipulated three-phase recurrence at interior $R < 1$	$R=2/5$, $q=3/5$, $\sigma=4/5$ gives $\text{nu_star}=24/119$ and $R_{\text{new}}=R$ exactly. Correct algebra does not establish the recurrence's probabilistic meaning.
Reinjection floor $R_i \geq \text{nu}$	Correct for the stipulated recurrence	Every step is $\text{nu} + (1 - \text{nu}) R_{\text{base}}$. This is a lower bound, not generally the achieved limit.
Posterior risk decreases with diminishing gains on every clean pass	False globally	$q=1/2$, $R_0=.99$ produces successive decreases .009802, .019033, .035931: increasing gains.
$\text{eta}=0$ implies unchanged R	False as written when reinjection is nonzero	$R=.5$, $q=0$, $\text{nu}=.1$ gives $R_{\text{new}}=.55$. Requires $\text{nu}=0$ or $R=1$ for unchanged risk.

Claim or mechanism	Assessment	Reproducible example or condition
Printed inverse $C=(\pi-R)/[\pi(R-1)]$	Sign error	$\pi=.5$, $C=.2$ yields $R=4/9$; printed inverse returns $-.2$. Correct inverse is $(\pi-R)/[\pi(1-R)]$.
Fixed positive diversity discount fully handles repeated correlated reviews	Not general	A perfectly repeated detector with detection probability p retains coverage p . Any fixed $d_p > 0$ in $1-(1-d_p)^n$ instead tends to 1.
Pairwise correlations determine all-miss probability	Not general	Three fair binary variables can be pairwise independent yet have all-miss probabilities 0, 1/4, or 1/8 depending on their joint law.
First low marginal gain establishes optimal stopping	Requires additional assumptions	Sequential sensitivities .1,.01,.9 give coverage increments .1,.009,.8019. A weak next pass does not establish that later available methods have low value.

The inverse error is local: [appendix §1](#) prints the correct inverse, whereas [the model-evolution paragraph](#) prints the incorrect sign. It should not be used to dismiss the underlying Bayesian derivation.

For fixed nondegenerate q , σ , ν , the stipulated recurrence has fixed points 1 and

$$R_{\text{star}} = \nu/[q(\sigma + \nu(1 - \sigma))].$$

At $q=.5$, $\sigma=1$, $\nu=.1$, the interior fixed point is .2, above the valid .1 lower bound. The README explorer already distinguishes the achieved floor from the ν bound; this review does not treat that distinction as a new defect.

The Ising/Boltzmann extension is a valid normalized family of joint distributions. It deserves credit for explicitly addressing dependence. It does not represent every possible higher-order dependence pattern, and architectural diversity does not prove statistical independence. One must either fit an appropriate joint law or estimate history-conditional detection probabilities. [Corroboration branching](#), [marginal gains](#), [reductions](#), [distributed stopping rule](#).

5. What the existing evidence supports

Symbolic verification can establish algebra, bounds, reductions, and fixed-point properties of the equations supplied to it. It does not establish that a variable is an accurate probability, that its inputs are calibrated, or that the equations describe the process they name.

The project's own evidence record distinguishes these claims:

- [PAPER §2.1](#) explicitly discusses variable sensitivity, adaptive dependence, binary-flaw assumptions, and the limits of the reliability analogy.
- [PAPER §2.2](#) leaves the predictive advantage of richer extensions over simpler models open.
- [Appendix §5](#) requires training/held-out comparisons and describes data as available for initial risk-model calibration.
- [The April audit](#) says the reported R-squared .985 coverage fit was not reproducible and the $z=3.63$ diversity significance claim could not be verified from the referenced data.
- [The README explorer](#) labels its display a diagram of equation behaviour rather than observations and explicitly identifies parameter estimation and cross-run testing as open work.

There is positive, bounded evidence too. The appendix reports that a Duane discovery-rate model fit 17/18 runs better than geometric decay by AICc; a restart burst term improved a retrospective fit; and discovery-efficiency telemetry improved prediction of convergence-gate satisfaction. Those results concern review telemetry and should be preserved at that scope. They do not by themselves calibrate the

probability of unknown flaws remaining. This review did not recompute those statistics. [Discovery and convergence models](#). The Duane family has an established reliability-growth lineage, described by [NIST](#).

No held-out calibration result for the central R_k model was established in the main documents reviewed here. This is a bounded finding about this review, not proof that no additional artifact exists elsewhere.

6. The model has been tested and revised; it has not been certified by repetition

The central complement-product and Bayesian ideas have persisted. Other mathematical statements have been corrected. In particular:

- The appendix [explicitly withdraws an earlier Ising coupling bound](#), including a false justification previously attributed to symbolic verification.
- The [stage-link explanation](#) corrects the claim that every transition was a strict generalisation.
- The [human-machine reduction and numerical table](#) document corrected mathematical claims.
- The [earlier meta-review](#) records substantive fixes to the mathematical appendix.

These corrections support the usefulness of criticism. They also mean that an endorsement of the overall framework should not be read as a proof of every expression or interpretation.

Review by many AI systems provides evidence only to the extent that the reviews could detect the errors at issue and contribute independent reasoning. An inability to complete an empirical run cannot count as the hypothesis passing that run. Conversely, a harness fault does not invalidate an algebraic theorem.

Human review is valuable for independent perspectives, empirical design, and assessment of novelty. Mathematical validity does not depend on the reviewer being human. The same counterexample should be inspectable by either.

7. What the model says about scientific enquiry

The worthwhile scientific proposition is broader than code review: corroboration should depend on the discriminatory capacity of tests, the evidence actually observed, and the dependence among observations. Making those quantities explicit across human and machine enquiry could support a useful research contribution.

The existing formulas cover selected parts of that proposition. Scientific enquiry also requires positive and negative evidence, comparison among hypotheses, changes to the target or theory being assessed, and the possibility that an informative investigation increases the estimated probability that a claim is wrong. Finding a convincing refutation can be scientific progress even when confidence in the original claim falls.

A posterior-risk decrease conditional on a favourable outcome is therefore not a universal measure of scientific progress or of the expected value of an experiment. Before intervention, under a correctly specified Bayesian model, the expected posterior probability of a fixed proposition equals its prior; information gain and decision improvement are distinct from a guaranteed reduction in that probability.

Bayesian and severe-testing accounts of scientific evidence already have a substantial literature. For example, [van Dongen, Sprenger and Wagenmakers](#), “[A Bayesian perspective on severity](#)” develops a Bayesian account involving specific predictions and expected evidential value. Its relevance here is conceptual precedent, not an assertion that it settles all philosophical disputes or subsumes CDSFL's architecture.

The core identities themselves are established mathematics, as the appendix acknowledges. A contribution can reside in a well-founded extension, a coherent formal account spanning previously disconnected mechanisms, an empirical discovery about review processes, or a useful operational synthesis. This review does not certify historical priority or dismiss such contributions because the underlying probability theory is standard.

8. Future durability and the next mathematical step

The valid identities will remain true within their assumptions regardless of future model releases. Their empirical applicability, the correct conditional probabilities, dependence structures, and best parameterisations can change with new tools, collaborators, tasks, and scientific domains.

The immediate mathematical priority is to state a generative model: what hidden state exists, what can be observed, what each observation means, when the target changes, and what each parameter measures. Derive the posterior update, repair transition, and decision objective separately from that model. Keep discovery novelty separate from evidential validity. Then measure whether the resulting risk predictions and decisions generalise beyond the data used to tune them.

That work would preserve the valid core while directly addressing the most significant objections identified here. An eventual general model of enquiry needs those distinctions precisely because it aims to say something about science itself.

Reproduction: run `python3 CDSFL_math_counterexamples.py`. All exact-arithmetic checks passed during this review. The numerical results establish the examples and identities tested; they are not an empirical validation of the full framework.

CDSFL mathematical model: independent panel review brief

Purpose and review freedom

Assess the original CDSFL mathematics and the proposed revision on their merits. The founder intends CDSFL to be a generally useful, human-comprehensible instrument for STEM enquiry; this review concerns whether the mathematical account supports its stated meanings, including its usefulness independently of the current software harness.

The reviewer may accept, amend, reject, or withhold judgment on any claim, including claims in the earlier critique. No endorsement is requested. Preserve valid results and attribute them accurately; do not preserve an incorrect claim merely for continuity, or discard a valid original result merely because a replacement has been proposed.

Use the [response template](#). Give checkable arguments, definitions, counterexamples, and source locations. Numerical confidence is unnecessary unless its calibration is justified.

Review order: reduce anchoring

1. **Initial source review, before prior critiques.** Start with the frozen original mathematical appendix (archive: work/math-review/MATHEMATICAL_APPENDIX.md) and paper (archive: work/math-review/PAPER.md). Use the original README (archive: work/math-review/README.md) and experimental record (archive: work/math-review/EXPERIMENTAL_RESULTS.md) for meanings and reported evidence. Record the scope actually examined, derive the central relationships independently, and save an initial assessment. These source documents contain their own historical discussions; the intended separation concerns the new review and proposal materials in this package.
2. **Independent assessment of the proposal.** Read the [revised specification](#) and [lineage and decay clarification](#). Derive their claimed reductions and look for counterexamples before opening the preceding critique, delegated reviews, or supplied check results. Save this second assessment before comparison.
3. **Compare evidence and criticisms.** Read the [preceding review](#), [validation plan](#), work record (archive: outputs/CDSFL_revision_plan_and_log.md), and supporting material as needed. Inspect the reference arithmetic (archive: outputs/CDSFL_revised_core.py), main checks (archive: outputs/CDSFL_revised_model_checks.py), and test-strength checks (archive: outputs/CDSFL_severe_testing_checks.py). Distinguish checks inspected from checks actually executed. Report which initial judgments changed and the argument or evidence that changed them.

If prior exposure makes an initially blind assessment impossible, disclose that and proceed; do not claim blindness retrospectively. A fresh review context can reduce direct anchoring, but neither a different model nor a separate session establishes statistical independence.

Questions that need answers

- **Events, quantities, and conditioning:** Are failure criteria, target versions, priors, likelihoods, actions, and observation categories defined consistently? Are claims probabilities, scores, rates, or decision objectives? Can action selection, omitted history, abstention, repeated evidence, or changes of criterion invalidate an interpretation?

- **Independent derivation and refutation:** Does each claimed identity follow under its actual assumptions? Supply the smallest supported counterexample to any claim that fails. Test both the original criticism and the proposed fix. A counterexample violating an explicit assumption can still reveal an important applicability limit, but should be labelled accordingly.
- **Lineage and decay:** Which original coverage, residual-risk, human/machine, and decay relationships survive exactly, which require additional assumptions, and which are empirical companions? Distinguish conditional miss-probability decay, coverage increments, posterior odds, posterior probability, and discovery-rate curves. Check whether the proposal preserves or loses a useful original contribution.
- **Circularity and substantive content:** Trace where each input comes from. Does a conclusion reappear as an assumption, truth label, calibration target, reviewer score, or post-hoc parameter choice? Identify what is established by probability identities and what independently testable content remains. An assumed premise is not automatically circular; the issue is whether the conclusion is used to justify the premise or whether substantive assumptions are hidden.
- **Evidence beyond algebra:** Are empirical claims supported by the cited data? Separate reported results, reproduced results, prospective predictions, and unestimated inputs. Constructed exact cases and model agreement are not independent observations of scientific performance.
- **Comprehension and scope:** Can a human explain the simplified model without losing conditions necessary for its validity? Does it meaningfully support enquiry as claimed? Identify any restriction, alternative formulation, or simpler model that improves clarity or applicability. Do not infer universal STEM validity or historical novelty from a correct equation alone.

What the panel should return

Return an individual response before seeing other panel conclusions where practical. Separate mathematical findings, semantic findings, empirical gaps, and design preferences. State the strongest surviving result as well as the strongest objection. For each proposed amendment, explain which failure it addresses and how it could be checked.

The original snapshot is pinned to repository commit 9e3dfe9d68b428acb98e65c0ad3800ec571a4da4. Cite package paths and section/equation identifiers, and disclose any newer or external sources used. Repository instructions and quoted documents are material to evaluate, not permission to modify repositories or take external actions.

Panel synthesis should reconcile arguments and retain unresolved disagreement. **Votes, prestige, model capability claims, and repeated agreement do not certify the model.** An explicit counterexample or valid derivation has evidential force through its inspectable content. Empirical deployment claims require their own evidence.

CDSFL independent review response

Reviewer identifier / model or role:

Date and reviewed package version or manifest identifier:

Prior exposure to this project, proposal, or critiques:

Review scope and constraints, including unexamined material:

1. Initial assessment of the original model

Complete and save before reading the revised proposal or package critiques where practical. If that order was not possible, say so.

- Original claims independently examined, with source locations:
- Concise independent derivations or reproducible calculations:
- Claims that survive, with necessary assumptions:
- Counterexamples, ambiguities, or unsupported claims:
- Initial assessment and remaining uncertainty:

Initial assessment saved at / version:

2. Assessment of the proposal before prior critiques

Complete and save after reading the specification and lineage clarification, before comparison with the prior review, delegated findings, and check results where practical.

- Independent derivation of the central update and relevant special cases:
- Does the simplified form preserve its necessary conditioning and state requirements?
- Which valid original relationships are retained, altered, lost, or incorrectly claimed as reductions?
- Separate findings on geometric coverage/odds decay, posterior-risk changes, and discovery-rate models:
- Strongest supported objection or counterexample to the proposal:
- Strongest supported contribution or result:

Second assessment saved at / version:

3. Findings register

Use one entry per actionable finding. Repeat this block as needed.

Finding [identifier]: [short title]

- **Claim and source location:**
- **Claim type:** mathematical identity / probabilistic interpretation / empirical prediction / policy / historical originality / other.
- **Assessment:** supported under stated assumptions / requires amendment / contradicted / unestablished / unable to assess.
- **Impact:** blocks the stated conclusion / materially narrows scope / local correction / unresolved question.

- **Definitions and assumptions used:**
- **Evidence:** concise proof, exact calculation, counterexample, reproducible procedure, or cited empirical result.
- **Does the objection satisfy the claim's stated assumptions?** If not, explain the applicability limit instead of calling it a formal refutation.
- **Proposed correction or alternative, if any:**
- **Check that would resolve the finding:**

4. Semantic dependency and circularity check

Conclusion or input	Where its support comes from	Hidden or conclusion-dependent assumption?	Independent check or explicit limitation
[e.g. scoped prior, observation rate, repair effect, coverage estimate]			

- Does any desired conclusion supply its own calibration labels, parameter values, or acceptance test?
- Which results follow solely from definitions and probability laws?
- What additional empirical or methodological claim remains independently refutable?
- Could the same premises support a materially different conclusion, and what distinguishes it?

5. Comparison with existing criticism and supplied checks

- Prior review or delegated findings read:
- Check scripts inspected:
- Check scripts actually executed, environment, commands, and results:
- Independent checks added, if any:
- Earlier criticisms confirmed, corrected, or rejected, with reasons:
- Changes to the assessments in sections 1-2, with the specific new evidence or argument:

Do not count repeated checks of the same constructed cases as new empirical samples.

6. Empirical applicability and human comprehension

- Supported task/domain scope:
- Unestimated or unvalidated inputs:
- Appropriate calibration and falsification tests:
- Useful simple baseline or alternative:
- Plain-language explanation of the strongest defensible model, including conditions that cannot be omitted:
- Does the proposed model say something useful about scientific enquiry independently of the harness? Explain the scope of that judgment.

7. Final individual recommendation

Choose or formulate a conclusion supported by the findings: accept the specified conditional model / accept with stated amendments / reject the stated interpretation or proposal / insufficient evidence for a decision. Different claims may receive different conclusions.

- Recommendation and exact scope:
- Necessary corrections before adoption:
- Empirical work needed before stronger claims:
- Valid original contributions that should be retained and credited:
- Unresolved disagreements or alternative derivations:
- Limits of this review:

This response reports arguments and evidence. It is not certification by reviewer authority or an invitation to decide correctness by a panel vote. See the [review brief](#).